



南京大學

$$1. p = \frac{W}{t} = \frac{N h \nu}{t} = n h \nu \Rightarrow n = \frac{p}{h \nu} = \frac{p \lambda}{h c} = 2.5 \times 10^{20} \uparrow$$

$$\sigma = \frac{n}{4 \pi r^2} = 2.0 \times 10^{19} \text{ 个/m}^2 \text{ 光子密度很大.}$$

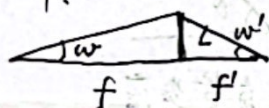
$$2. \tan \frac{\theta}{2} = \frac{D/2}{d} \text{ 由 } \theta \rightarrow 0, \text{ 得 } d = \frac{D}{\theta} = \frac{2.5 \text{ cm}}{10^{-3} \times \frac{1}{360} \times \frac{\pi}{180}} = 5.2 \times 10^8 \text{ cm} = 5.2 \times 10^3 \text{ km.}$$

(注 d 为距离)

$$3. \text{ 由费曼车锁 } m = 2.1 + 5 \lg D(\text{mm}). \text{ 得 } m = 2.1 + 5 \lg (2.4 \times 10^3) = 19 \text{ mag.}$$

$$\text{由瑞利判据 } \theta = \frac{1.22 \lambda}{D} \text{ 望远镜口径时 } \theta = \frac{110''}{D(\text{mm})} = \frac{110''}{2.4 \times 10^3} = 4.6 \times 10^{-2}''$$

$$4. \tan W = \frac{L}{f} \quad \tan W' = \frac{L}{f'}. \text{ 要求 } W' > 1'.$$

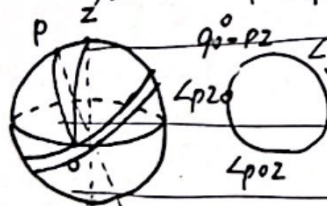


$$\text{则 } f' \leq \frac{L}{\tan W'} = \frac{16.7 \times 10^{-4} \text{ cm}}{1' \times \frac{1}{60} \times \frac{\pi}{180}} = 5.7 \text{ cm.}$$

使用 1.25 cm, 2.5 cm, 5 cm 目镜可以.

$$5. \text{ 由费曼车锁 } m = 2.1 + 5 \lg D(\text{mm}). \text{ 得 } m = 11 > 10.75 \text{ mag.}$$

考虑该天体落子. 球面三角关系 pzo . $\angle zpo = t$.



$$\text{直边三角关系 } zo = 90^\circ, pz = 90^\circ - \phi, po = 90^\circ - \delta$$

$$\begin{aligned} \sin(t - 90^\circ) &= \tan(90^\circ - (90^\circ - \phi)) \tan(90^\circ - (90^\circ - \delta)) \\ -\cos t &= \tan \phi \tan \delta \Rightarrow t = 91.4^\circ = 6.2 \text{ h} \end{aligned}$$

(不考虑大气消光)

$$6. \text{ 由 } d \sin \theta = \lambda m. \quad d \cos \theta d\theta = m \lambda d \Rightarrow d\theta = \frac{(d \lambda) m}{d \cos \theta} \approx \frac{\lambda m}{d}$$

$$\text{当 } m=2, \lambda = 0.597 \text{ nm}, d = \frac{1}{300} \text{ mm}, \text{ 得 } d\theta = 3.58 \times 10^{-4} \text{ rad}$$

$$\frac{\Delta}{\Delta \lambda} = N m \quad \text{取 } \lambda = \frac{\lambda_1 + \lambda_2}{2} = 589.589296 \text{ nm} \quad N = \frac{0.597}{589.296 \times 0.517 \times 2} = 494.2$$

$$\lambda = 589.594 \text{ nm.}$$

最大波数. 保证 N 是最大值.

$$\frac{589.594}{0.597 \times 2} = 494.2$$

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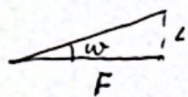
8.

$$n = 2048 + 2048 = 4096 \text{ 次}$$

7. 最大误差的像素点所在电荷一共转移 $(2048-1) + (2048-1) =$

$$\text{最大误差} = \frac{1}{(1-0.9999\%)^{4096}} - 1 = 0.0(4.0\%)^{4096} =$$

8. 由 $D = 4.0 \text{ m}$, $F/\phi = 3.0$, $F = 12 \text{ m}$.



$$L = F \tan w = 12 \times 0.6'' \times \frac{1}{3600} \times \frac{\pi}{180} = 9.5 \times 10^{-5} \text{ m} = 95 \mu\text{m}$$

当 CCD 像素尺寸 $> L$ 时, 可消除视宁度的影响

故像素尺寸为 $35 \mu\text{m}$

由 $D = 15 \text{ cm}$, $F/\phi = 3$, 得 $F = 45 \text{ cm}$.

$$L = F \tan w = 45 \times 0.6'' \times \frac{1}{3600} \times \frac{\pi}{180} = 1.3 \times 10^{-4} \text{ cm} = 1.3 \mu\text{m}$$

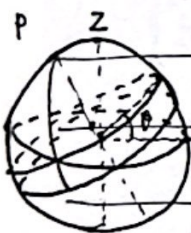
像素尺寸为 $1.3 \mu\text{m}$.

5. 讨论天体高度角变化对于地面辐射接收的影响

天体位于天顶时 $f = f_0 \cos \theta_{\text{max}}$, f_0 是垂直接收时的辐射通量密度

当望远镜处于极限位置时 $f' = f_0 \cos \theta$

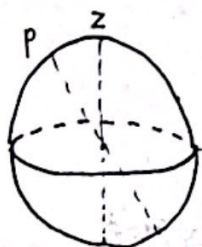
$$m - m' = 2.5 \lg \frac{f'}{f} \quad m = 10.75 \text{ mag}, \quad m' = 2.1 + 5 \lg D (\text{mm}) = 2.1 + 5 \lg 65 = 11.16 \text{ mag}$$



$$\theta = \epsilon + \delta = 32^\circ + 5^\circ 27' = 37^\circ 27'$$

ϵ 是当地纬度

$$\frac{f_0}{f} = 0.41 = 2.5 \lg \frac{\cos \theta}{\cos(37^\circ 27')}$$



5. 设消光系数为 k .

地理纬度取 $\delta = 32^\circ$.

当恒星上中天时, 高度角 $h = 63^\circ 27'$, 时间 $t_0 = 0^h$.

望远镜极限星等 $m' = 2.1 + 5 \lg D = 11.16 \text{ mag}$.

由大气消光, $m' = m_0 + \frac{k}{\cos z}$

将地平坐标系转换为时间坐标系, 得: $\begin{cases} \cos \delta = \sin \gamma \cos z + \cos \gamma \sin z \cos A \\ -\sin \delta = \sin \gamma \sin z - \cos \gamma \cos z \cos A \end{cases}$

联立解得 t' 当 t' 后无法观测.

